

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

CONSTRUCTOR GROUP *Machine Learning Consultant* 2021–present

NATIONAL UNIVERSITY OF SINGAPORE *Postdoc under [Kostya Novoselov](#)* 2022–present

Wyckoff Transformer for generating symmetric crystals [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

ML for defects in 2D crystals [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference: 4x less memory, 8x fewer GPU operations
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

CERN [YANDEX → HSE UNIVERSITY] *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

CatBoost aka fighting biases with dynamic boosting [paper](#)

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

EDUCATION

Higher School of Economics (HSE University) <i>PhD in Computer Science</i> , supervisor Andrey Ustyuzhanin ; Machine Learning for particle identification in the LHCb detector .	2016–2020
Sapienza – Università di Roma <i>PhD in Physics</i> (double degree with HSE), supervisor Barbara Sciascia .	2016–2020
Product Management course at Yandex Focused coursework in product strategy and execution.	2018
Yandex School of Data Analysis Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> (2014–2016): Optimisation of data processing of the LHCb experiment. <i>BS in Physics</i> (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.	
International Junior Science Olympiad (IJSO) Silver medal (held in Korea).	2008

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).

TEACHING & OUTREACH

- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **C++** – Production
- **Linux** – Administration
- **Physics** – Deep domain understanding

SELECTED PUBLICATIONS

1. Kazeev, Nikita, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
2. Kazeev, N., Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).
3. M. Borisyak and N. Kazeev. [Machine Learning on data with sPlot background subtraction](#). Journal of Instrumentation 14.08 (2019). **Alphabetic order, I'm the corresponding author.**
4. Derkach, D., Kazeev, N., Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. **Alphabetic order, I'm the corresponding author.**